توجيه الكمبيوتر وتكنولوجيا المعلومات

سفراء التطوير

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البرمجه و الذكاء الاصطناعي

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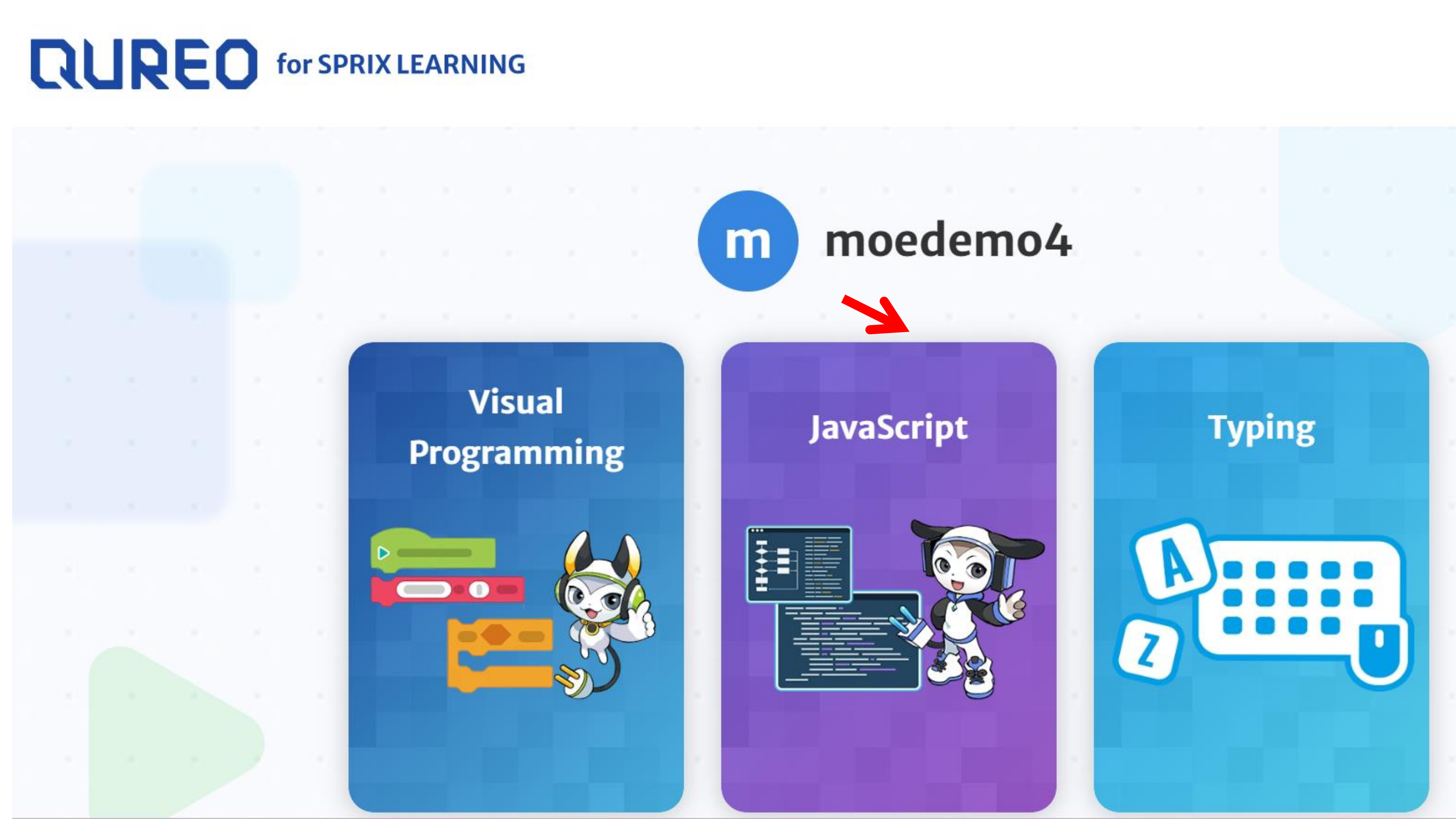

QUREO for SPRIX LEARNING
m
moedemo4
Visual Programming
JavaScript
Typing
A
Z

QUREO for SPRIX LEARNING

Acquired diamonds 5 / 150

Chapters completed 1 / 30

START
learning

Try out what you've learned

Open Freely Editor

Chapter List

◆ PERFECT

1 Introduction

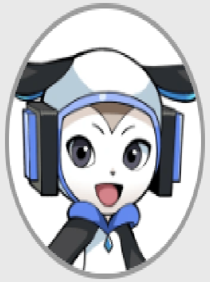
2 Calculations and Strings

3 Variables

4 Comprehension Check

5 If statement

[Chapter1]Introduction



Welcome to the world of programming! In this chapter, we will have fun learning programming together. We will start by asking, "What can programming do?" and progress to writing your own program to display text on the screen.

1	What is programming? (i)	2	What is programming? (ii)
3	How to use an editor	4	Console

1 What is programming? (i)

2 What is programming? (ii)

3 How to use an editor

4 Console

5 Comment

6 Comment out

Review

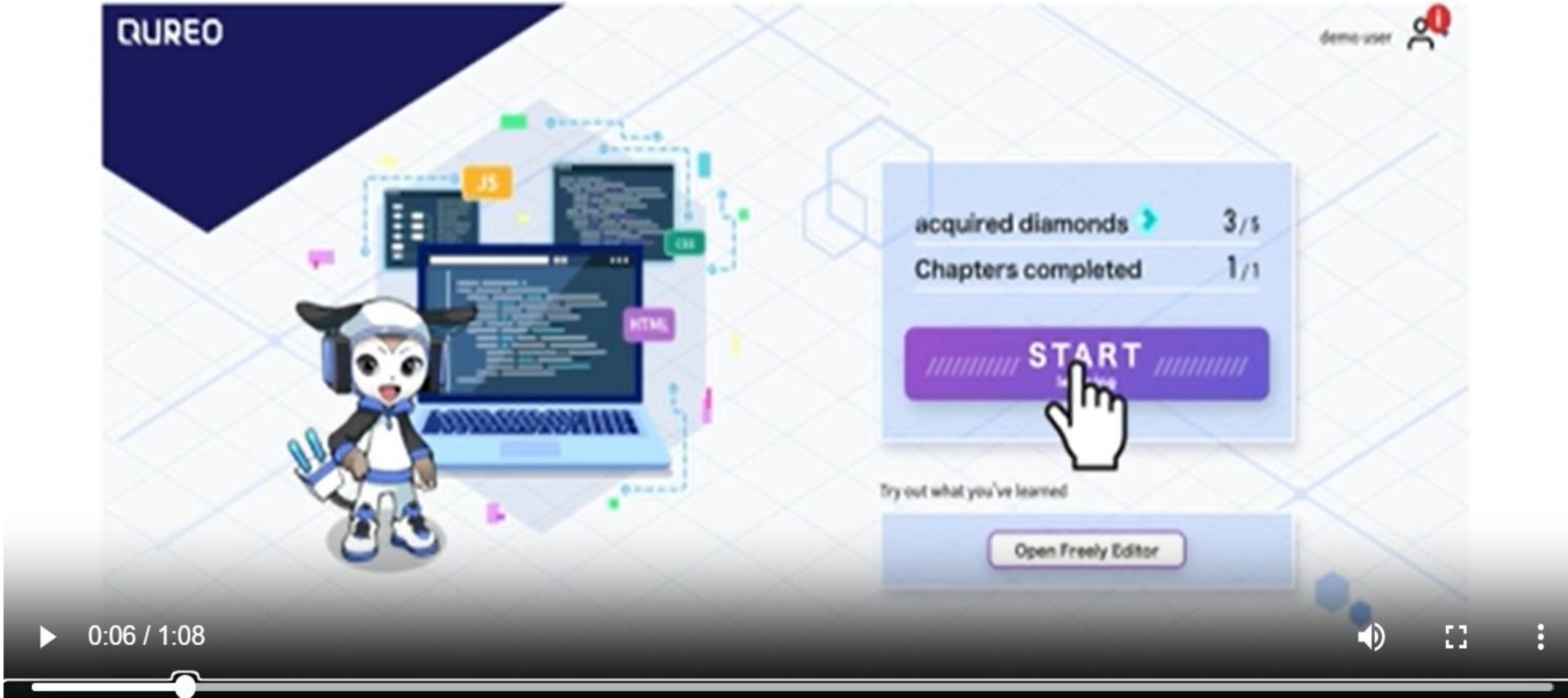
[Lesson1]What is programming? (i)

Key Points of the Lesson

What you can do with programming


Text Programming Course
How to proceed
with the lesson
0:03 / 1:08

Start!

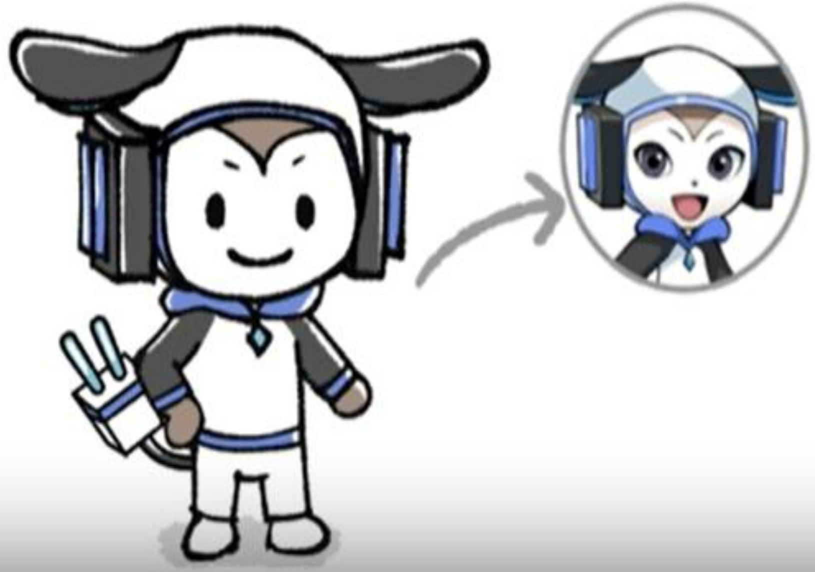

QUREO
demo user
JS
CSS
HTML
acquired diamonds 3/5
Chapters completed 1/1
START
Try out what you've learned
Open Freely Editor
0:06 / 1:08

Student **Rick**

A friend who learns programming together.

Teacher **Cody**

They teach us programming.

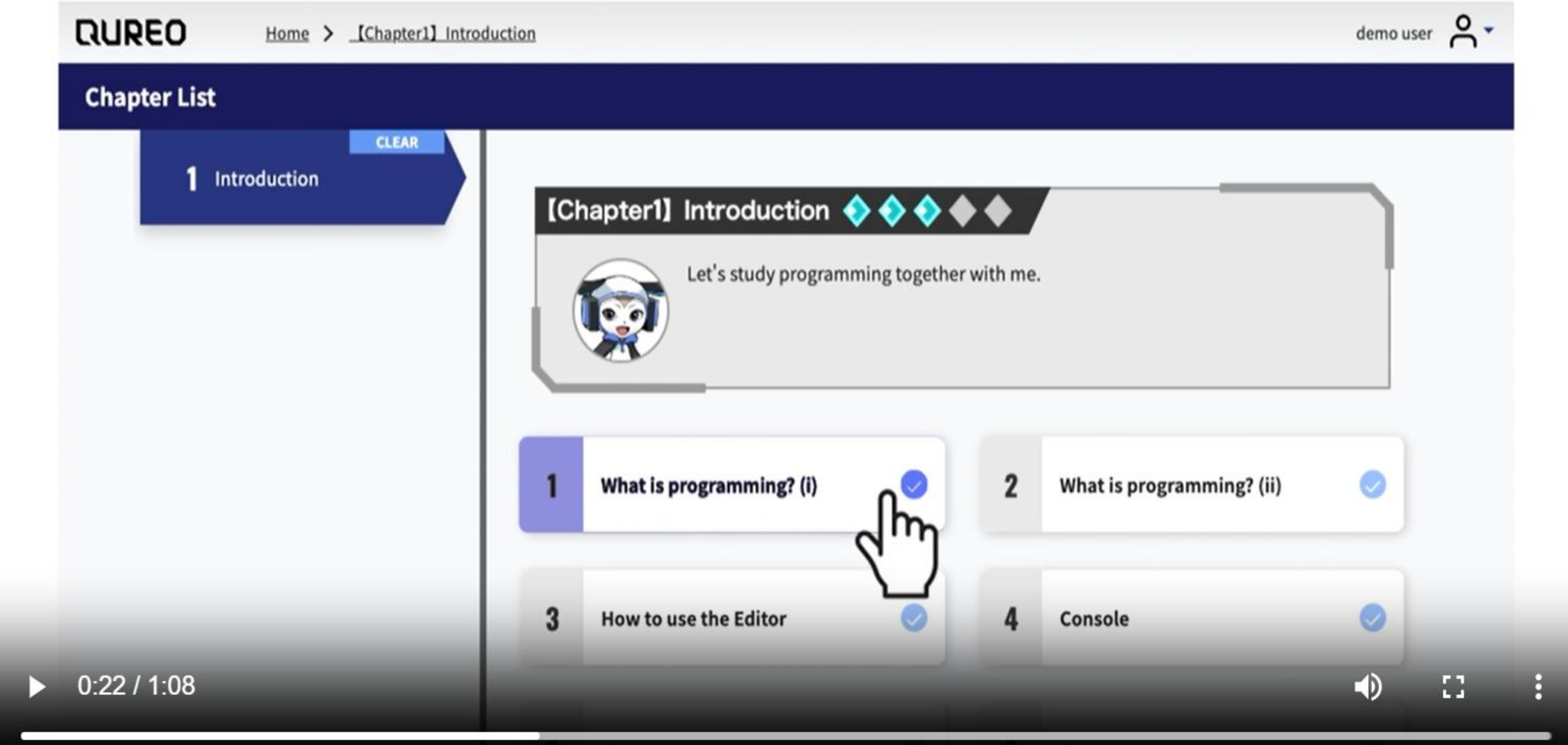


0:14 / 1:08

You two will be **learning together.**

0:16 / 1:08

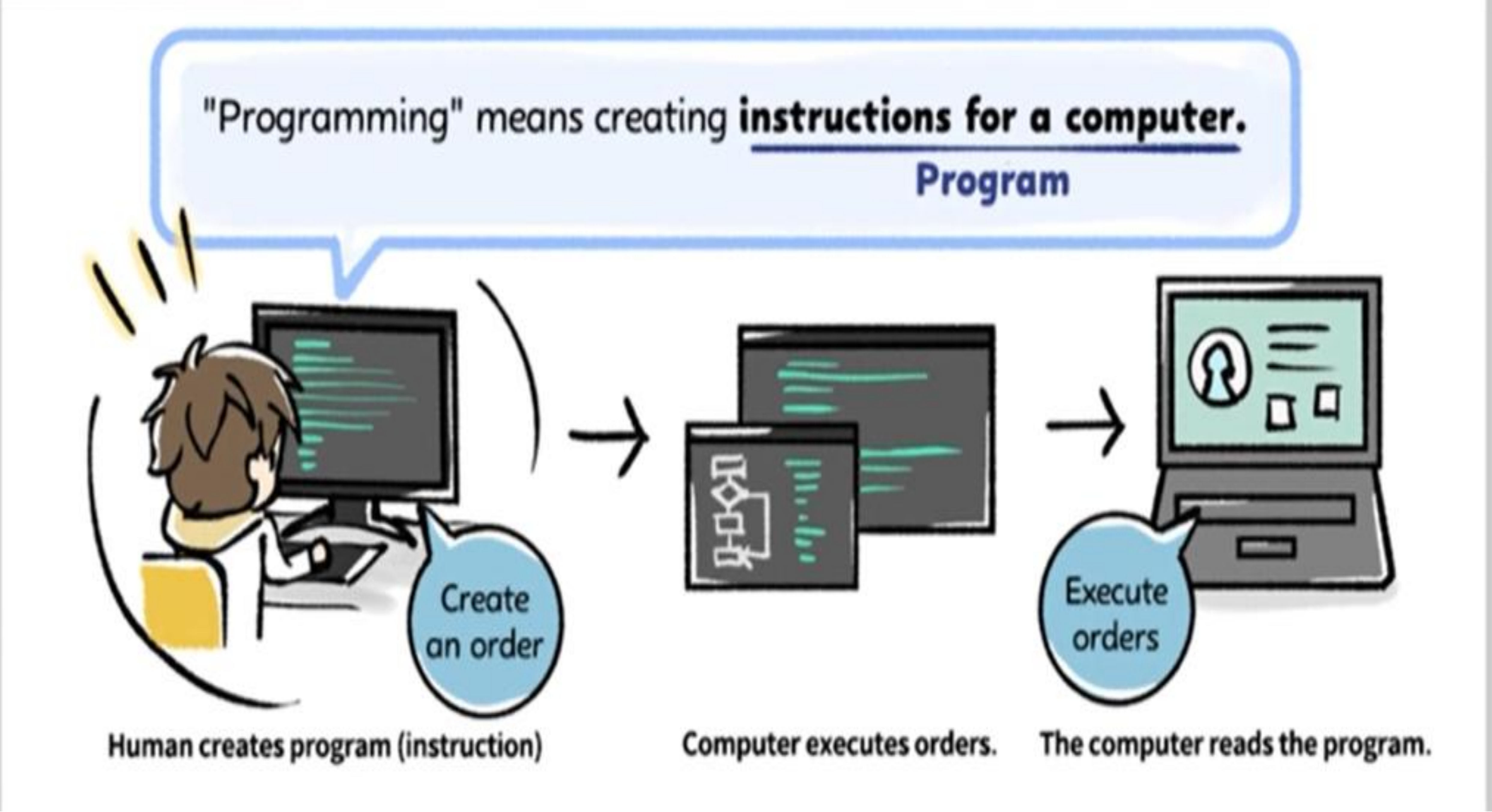
Choose a lesson



First,
check what you will learn in this lesson!

The Lesson's Key Points

- What is programming?
- What works with programming
- Types of Programming Languages


"Programming" means creating instructions for a computer.
Program
Create an order
Execute orders
Human creates program (instruction)
Computer executes orders.
The computer reads the program.

Can you start a computer by writing a program?
Also, did the robot move because you wrote a program and gave an instruction to it?

Exactly! The Robot also has a computer inside it, so speaking to the computer, in other words, writing a program, made the Robot move.

Now for the second question!
There are many things other than robots that are moved by programming!
What do you think?

You mean something with a computer inside... Hmmm, what is it?

Actually, various products around us, such as personal computers, home appliances, and automobiles, are moving by program.

Things around you where the program is used

Whoaa! Programming is used in our immediate surroundings.

That's right! This is how a lot of different products and services are created to make society more convenient.

Now, the last question!
We already mentioned that we use a special word that computers can understand called a "programming language" for programming.

How many programming languages are there in the world?

Eh ~? There's only one programming language, right?!
Hmmm, I don't know. Answer instead!

Teacher

Cody

He is the teacher who will teach you and Rick the programming.
He seems to create various games and apps.

Student

Rick

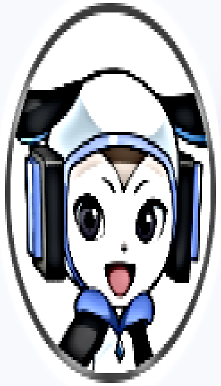
He is a friend who learns programming with you.
He is a mood maker who likes to play games.

Question

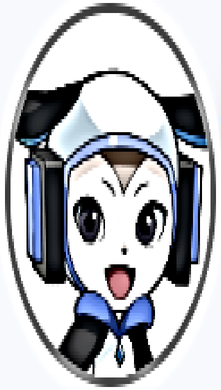
(Join Rick for a conversation! Answer Rick's questions.)
Let's play on the computer later!

- I want to play with it!
- I don't want to play.

Wow, that's great! We want to make one too!
But isn't programming difficult...?

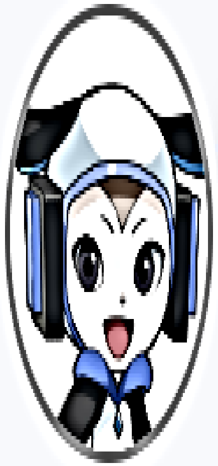


Some parts are difficult, but it's okay if you start with the easy parts!
It's a lot of fun, so let's do it together!

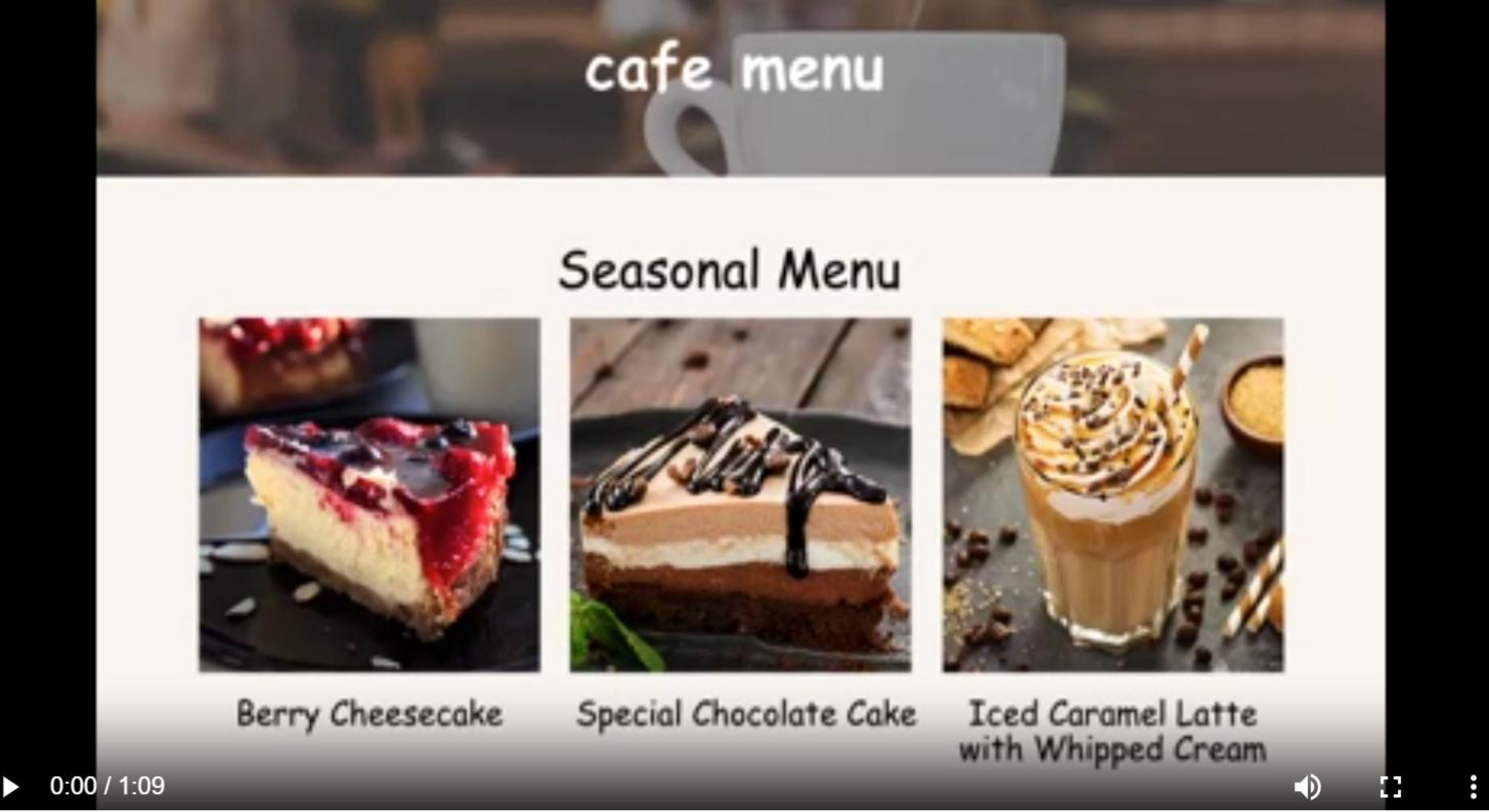


By the way, if you can do programming, you can make not only games, **you can also make a lot of other things.**

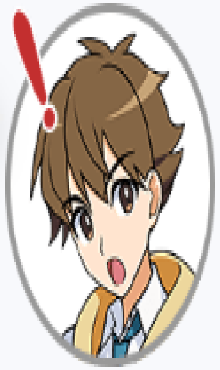
Other things? Like what for example?

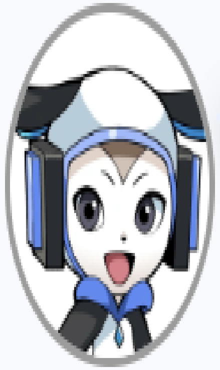


You know that cafe that recently opened in front of the station? Actually, I also created the website of that cafe!


cafe menu
Seasonal Menu
Berry Cheesecake
Special Chocolate Cake
Iced Caramel Latte
with Whipped Cream
0:00 / 1:09

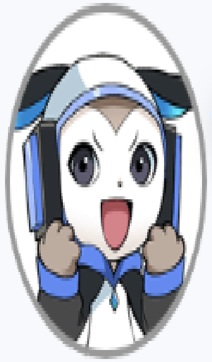
What, Cody made that store's website? That's amazing...
I wonder if I can make my own website too?





It's good that you have something you want to make!
When you can program, you can even make things like this.

What? A chat app? I use one every day!
I can't believe we can make something like that! Programming is amazing!



Are you starting to get a little interested?
I'll teach you and we'll learn how to program together and make all kinds of things!

Yes, I'd like that! Cody, I look forward to learning how to program with you!

Check the Key Points

Let's recap what you have learned in this lesson. Check the items below and click the "Got it" button.

- [x] Programming allows you to create games, website, apps, etc.

Got it

Back to Chapter List

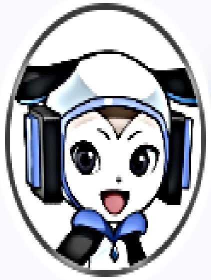
Continue to Next Lesson

Key Points of the Lesson

What is programming?

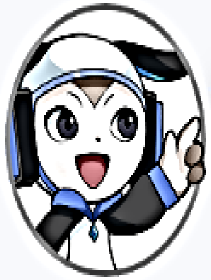
Things that work with programming

Types of programming languages



Now, before we start programming... First, a quiz to get you ready!

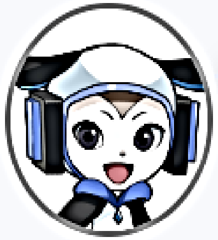
I don't know anything yet, but I wonder if I can answer the questions...



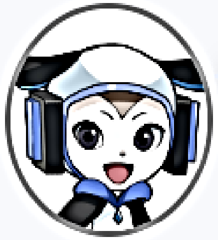
Now the first question!
What exactly is programming?

Huh? Hmmm, you said earlier that you can create games and apps through programming, right?

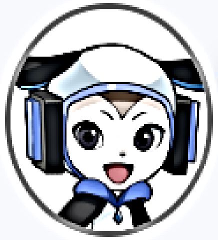
Oh! Come to think of it, there may have been times in school when we put blocks together to make games or move robots...
Was that Programming too?



Oh! You've done some programming.

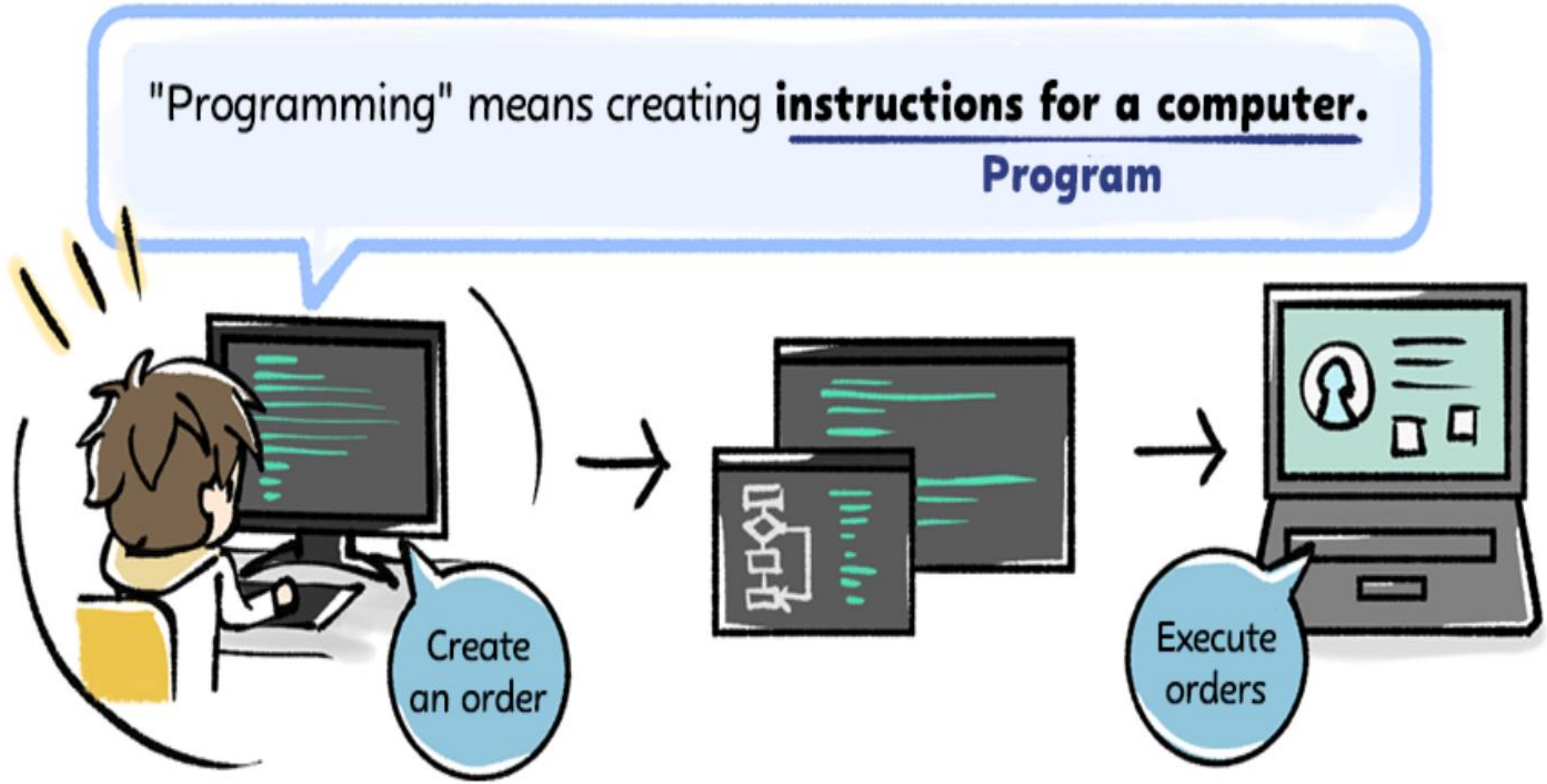


"Programming" simply means giving instructions to a computer.



Think of it like writing an “instruction manual" that uses special words that computers can understand, called **“programming languages"**, to give instructions to the computer.
These instructions are called a **“program"**.

What is programming?



Human creates program (instruction)

Computer executes orders.

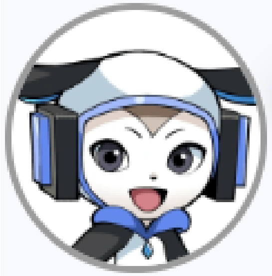
The computer reads the program.

If I write a program, can I make a computer work?
Does it mean that the robot moved because I wrote a program and told it to move?

Exactly! The robot also has a computer inside it, so you gave instructions to that computer, or in other words, you wrote a program that made the robot move.

Now for the second question!
There are lots of things that are controlled by programming, not just robots!
What do you think they are?

You mean something that has a computer in it... Hmm, what could it be?



In fact, **many of the products around us, such as computers, home appliances, and automobiles, are programmed.**

Familiar objects in which programs are used

Wow! programming is used in our daily lives, isn't it?

That's right! That's how various products and services are created and society is made convenient.

Now, the final question!
We talked about programming using a "**programming language,**" a special language that computers can understand.

How many programming languages are there in the world?

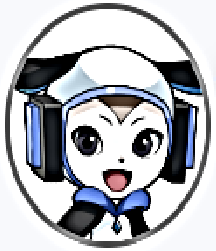
What? There's more than one programming language?!
Hmm, I don't know... Can you answer it for me?

Question

How many programming languages are there in the world?

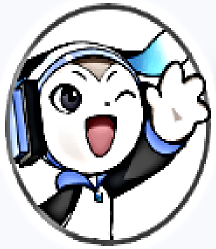
- 3 types
- 10 types
- More than 100 types

More than 100 types? I can't remember that many...
Cody, there's no way it's really 100, right?



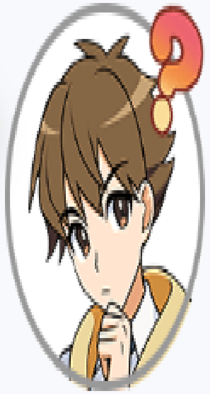
The correct answer is... **More than 100** programming languages!

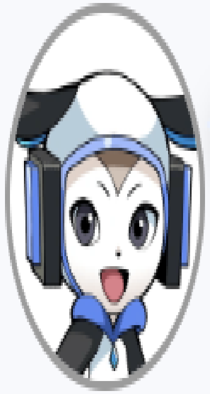
What! That many?
I'm going to have to learn more than 100 programming languages to make games and other works?



Don't worry, because we will be learning about three different programming languages together!
They are **"HTML," "CSS," and "JavaScript"**.

But why do we use three different types?

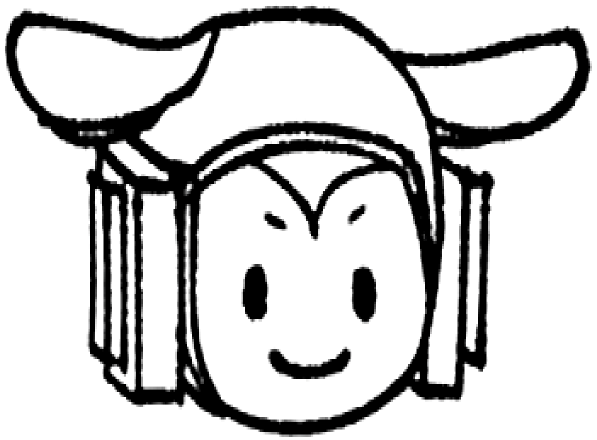




Different programming languages are good at different things.

Therefore we use a combination of them!

Differences in language



HTML

Build the framework

HTML

CSS

Create appearance

HTML + CSS

JavaScript

Create movement

HTML + CSS + JavaScript

※ HTML" is a markup language and "CSS" is a stylesheet language, which are not included in "programming languages.

I see that each of the three languages has its own role to play. I can't wait to write my own programs!

Not so fast!
Let's start by learning “JavaScript" in the next lesson!

"JavaScript," huh?
Alright! I'll do my best!

Summary of this lesson

- "**Programming**" is creating **instructions to a computer.**

  Program

- Programs are used in a wide variety of products around us, such as computers, home appliances, and automobiles.

- There are more than 100 programming languages, and here you will learn three types: HTML, CSS, and JavaScript.

Check the Key Points

Let's recap what you have learned in this lesson. Check the items below and click the "Got it" button.

- "Programming" means giving instructions to a computer.
- Programs run the various products that surround us, such as personal computers, home appliances, and automobiles.
- There are more than 100 different programming languages, and here we will learn three of them: HTML, CSS, and JavaScript.

Got it

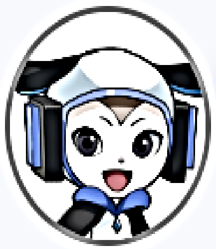
Back to Previous Lesson

Continue to Next Lesson

Key Points of the Lesson

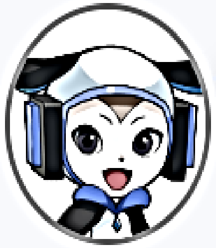
About text editors

How to type into a text editor



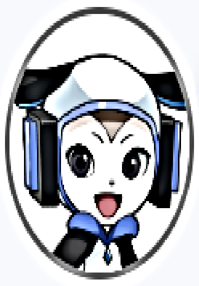
I hope you now have some idea of what programming is all about.

Yeah, I'm starting to understand it somewhat! I can't wait to try programming!



Well then, let's write a program!

Programs are typed on this black screen. This black screen is called a "**text editor**". It is also called an "**editor**" for short! You will use it often from now on, so please keep it in mind!

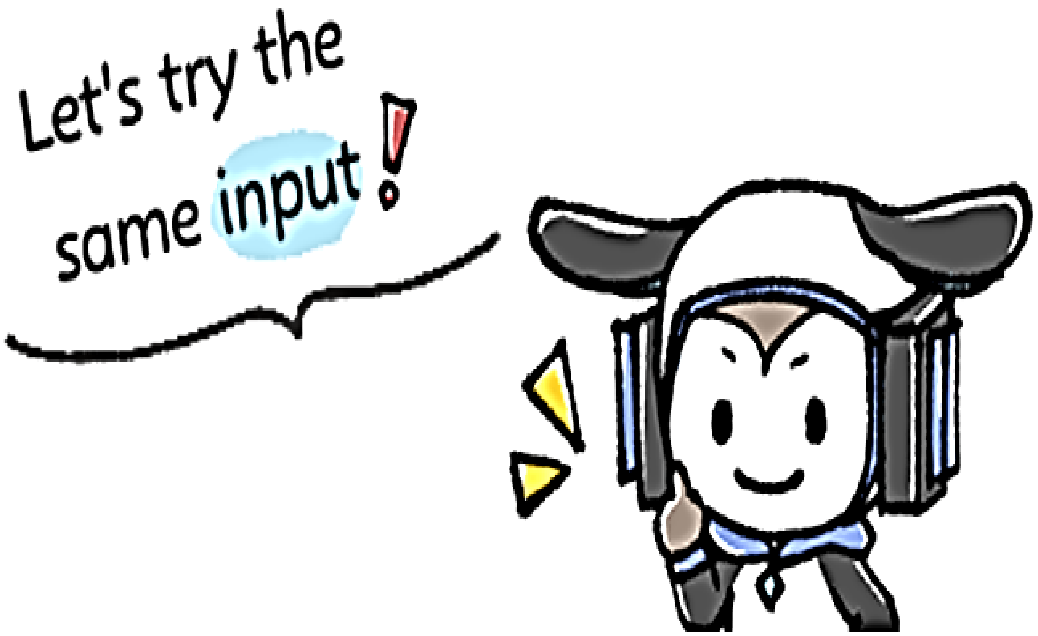


Let's start by typing "Hello" into the editor.

Typing into an editor

JavaScript

```
Hello
```



H...e...l...l...o... Done!

OK! But if you just write ordinary characters, the computer won't understand them.

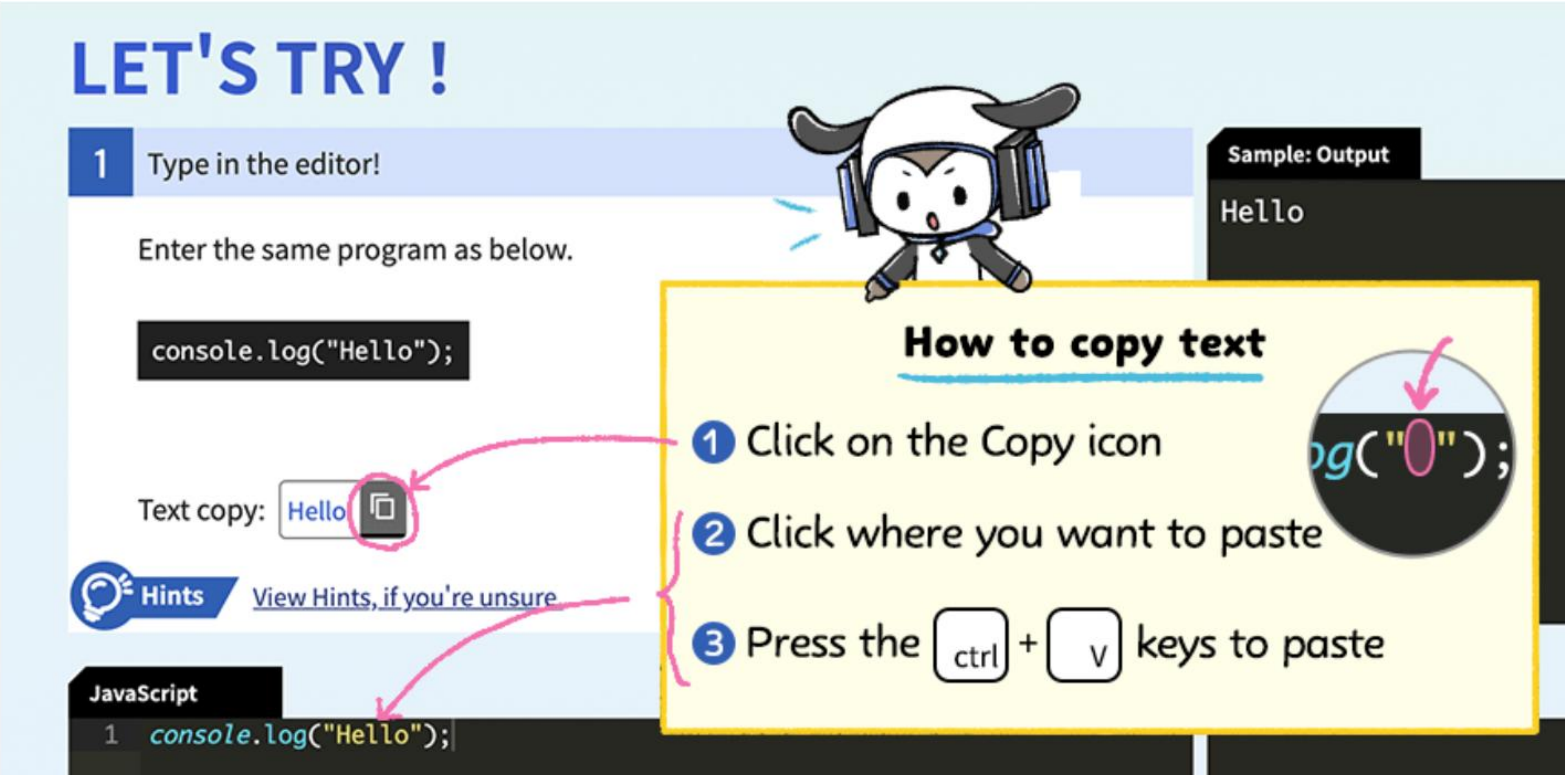
Next time, write it like this!

Typing into an editor

JavaScript

```javascript
console.log("Hello");
```

How to copy and paste text



LET'S TRY!

Try again

1 Type in the editor!

Enter the same program as below.

```
console.log("Hello");
```

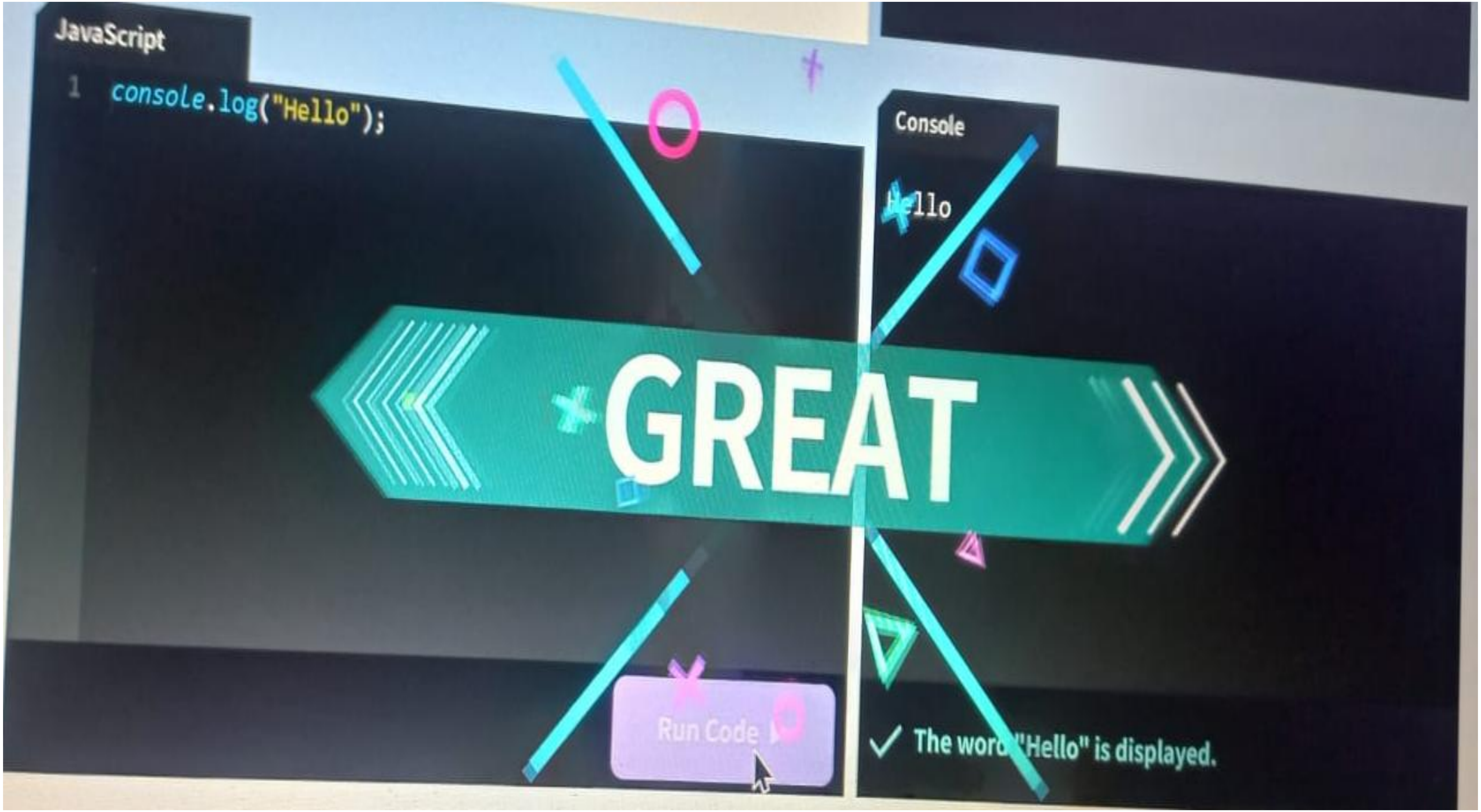
Text copy : Hello

Hints View Hints, if you're unsure.

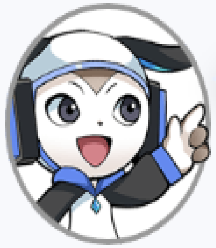
Sample: Output

```
Hello
```


JavaScript
1 console.log("Hello");
Run Code
Console
Hello
The word "Hello" is displayed.

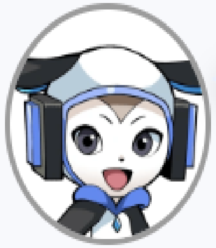

JavaScript
1 console.log("Hello");
Console
Hello
GREAT
Run Code
The word "Hello" is displayed.

The word “Hello" appeared on the screen!



Perfect! This part written in English and with symbols is the part written in the programming language.
If we don't write in a programming language, the computer can't understand our instructions.

You know, that “special language that computers can understand" that we learned in the last lesson!



Next time you can type it all up yourself.

LET'S TRY!

Try again

2 Type in the editor!

Enter the same program as below.

```
console.log("Good morning");
```

Text copy : Good morning

Hints View Hints, if you're unsure.

Sample: Output

```
Good morning
```

JavaScript

```javascript
consol.log("Good morning");
```

Console

```
Uncaught ReferenceError: consol is not defined
```

Reset the code

Run Code ▶

✗ Let's review the code

View Answer

JavaScript

```javascript
console.log("Good morning");
```

Run Code ▶

Console

```
Good morning
```

✓ The words "Good morning" are displayed.

Summary of this lesson

- The editor can also be used to write programs.
- Programs are written in **programming language**, not in ordinary letters.

JavaScript

```
console.log("Hello");
```

Check the Key Points

Let's recap what you have learned in this lesson. Check the items below and click the "Got it" button.

- [x] The place where you write your program is called a text editor (editor).
- [x] Programs are written in programming languages, not in ordinary characters.

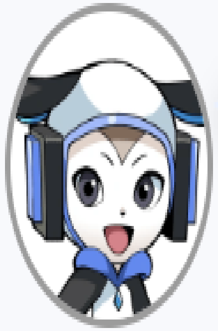
Got it

Back to Previous Lesson

Continue to Next Lesson

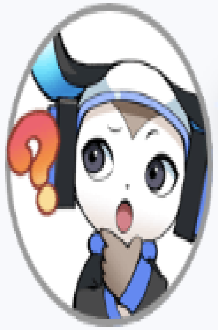
Key Points of the Lesson

- About consoles
- How to display text in the console
- About errors



Last time, you wrote a program using a programming language for the first time.

Yep! I was able to display words like "Hello" and "Good morning"!

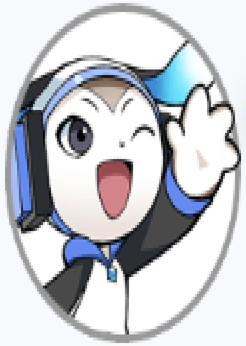


Do you remember what the black screen where you type in your program is called?

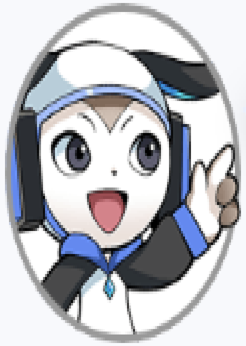
Question

What do you call the black screen where you type your program?

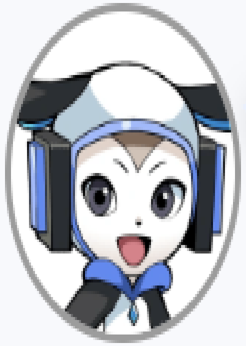
- Editor
- Blackboard
- Notepad



Great answer! You remembered it perfectly!



By the way, I didn't tell you this last time, but this black screen on the right is called a "**console**".



Today, let's learn a program to display text on the console!

LET'S TRY!

Try again

1 Type in the editor!

Enter the same program as below.

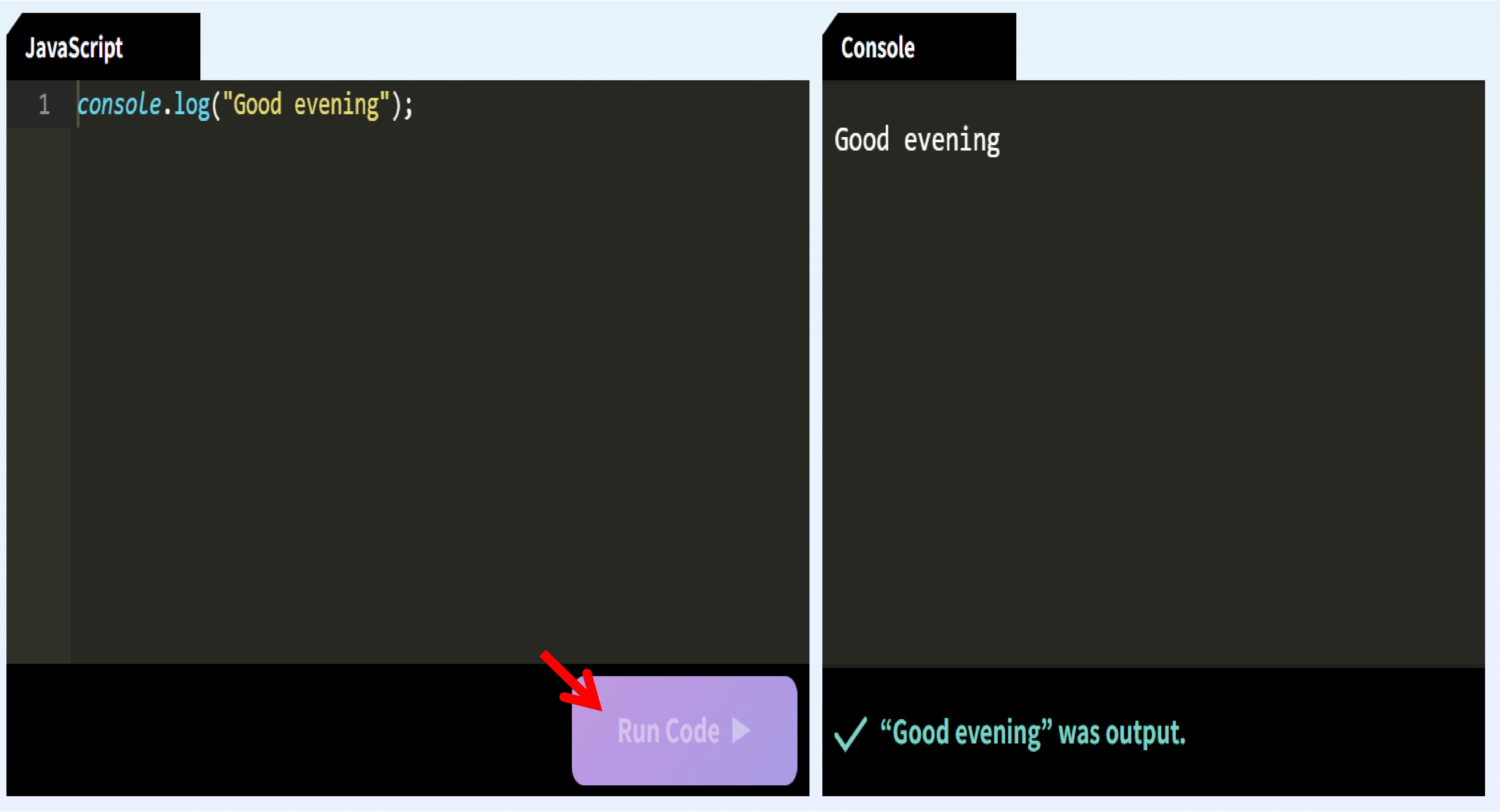
```
console.log("Good evening");
```

Text copy : Good evening

Hints View Hints, if you're unsure.

Sample: Output

```
Good evening
```

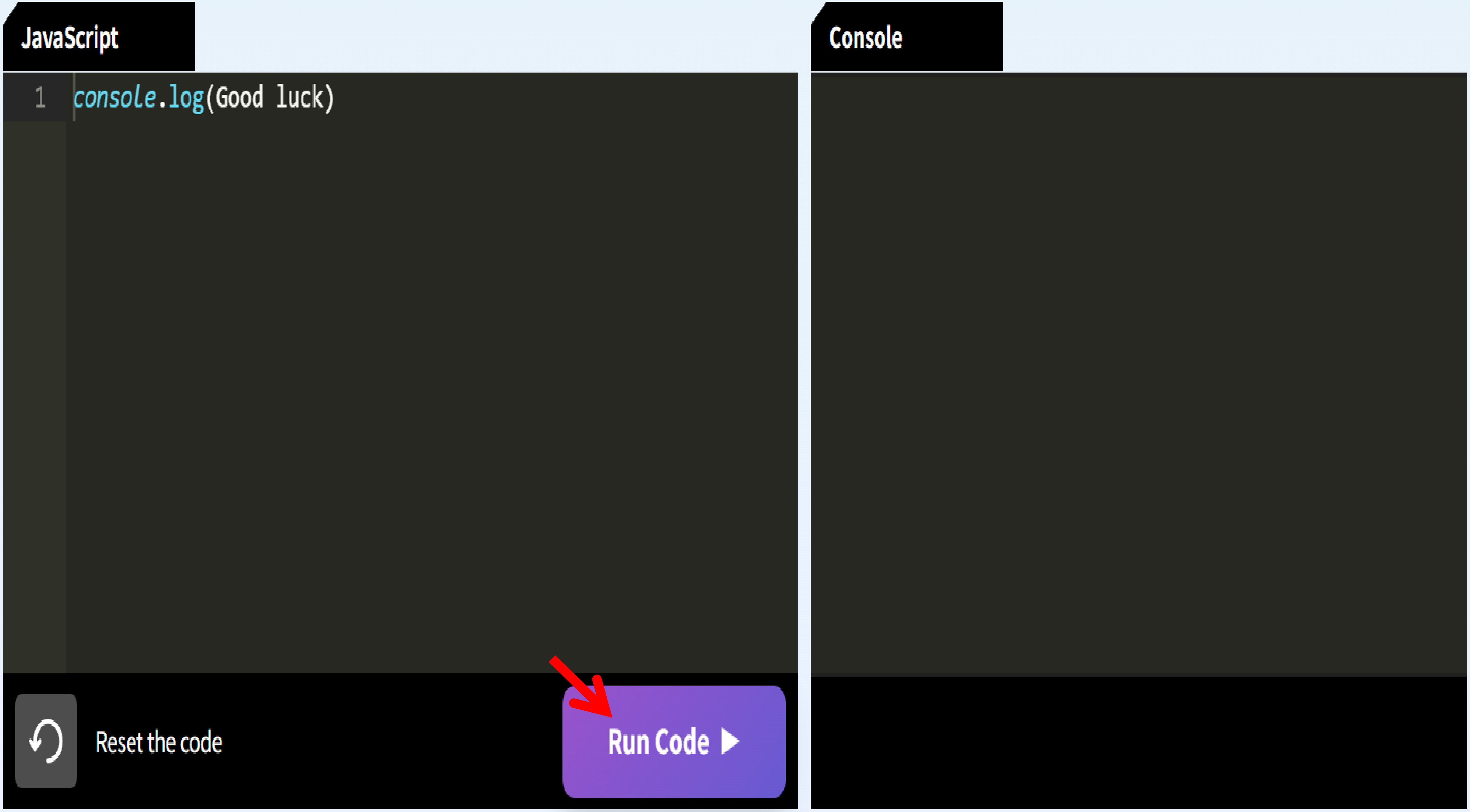

JavaScript
1 console.log("Good evening");
Run Code
Console
Good evening
“Good evening” was output.

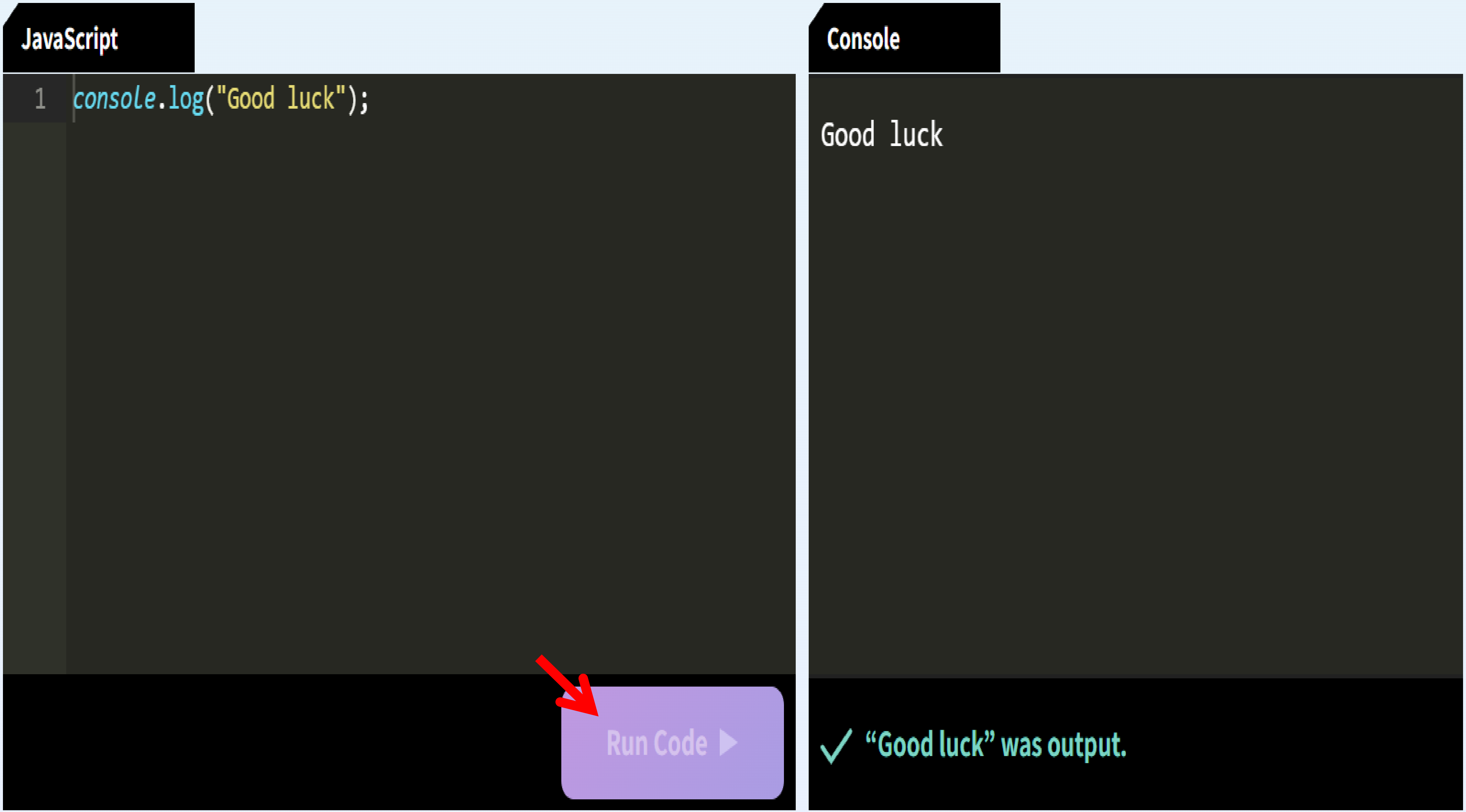
How to output to the console

JavaScript

```javascript
console.log("Hello");
```

Characters enclosed in
" (double quotation marks)
will be output to the console.


JavaScript
Console
console.log(Good luck)
Reset the code
Run Code


JavaScript
console.log("Good luck");
Run Code
Console
Good luck
✓ "Good luck" was output.

Summary of this lesson

- The input program can be executed and output to the console.
- Writing **console.log("~~")** will output the character ~~ in (" ").

JavaScript

```javascript
console.log("Hello");
```

Console

```
Hello
```

Check the Key Points

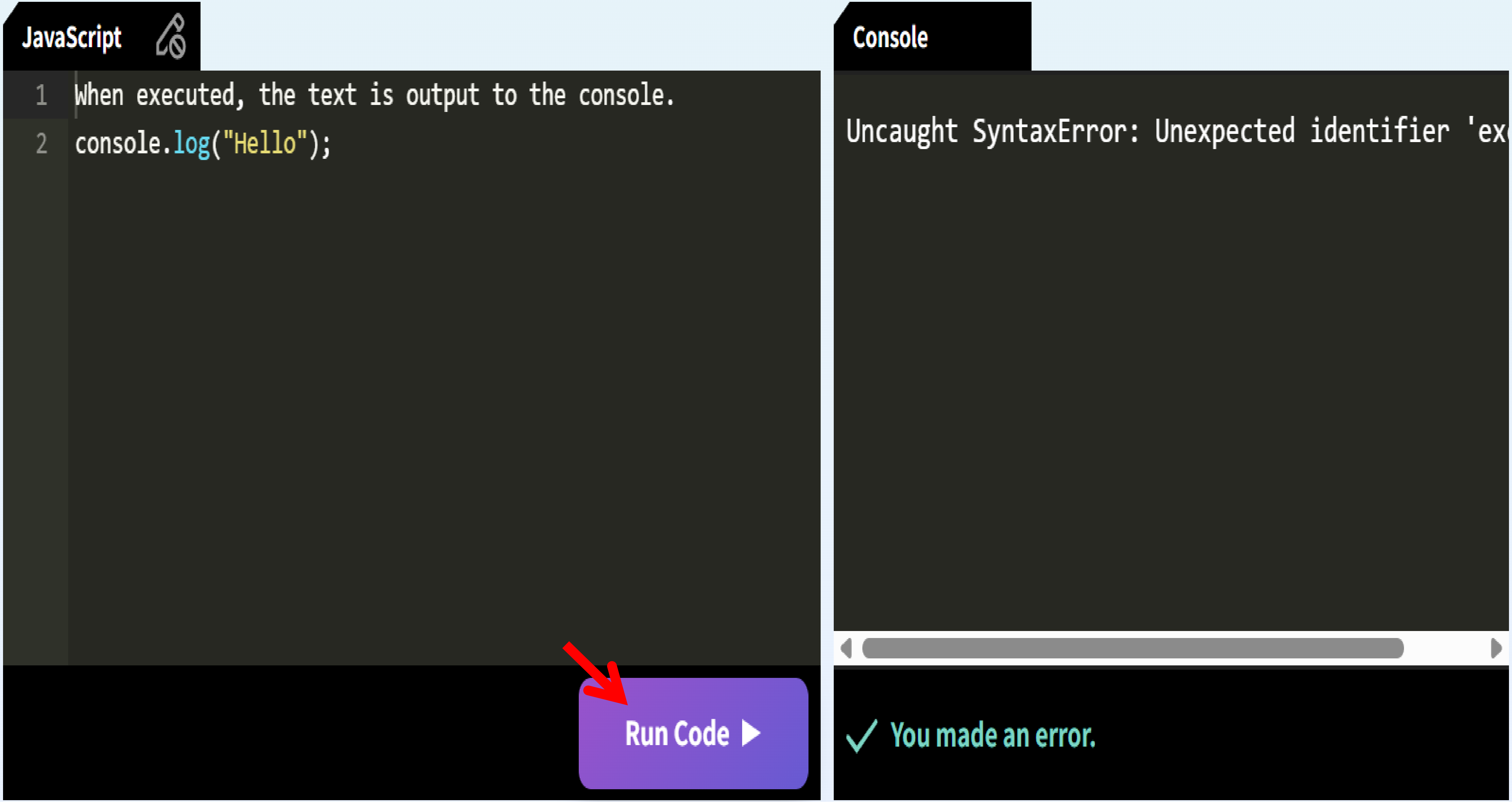
Let's recap what you have learned in this lesson. Check the items below and click the "Got it" button.

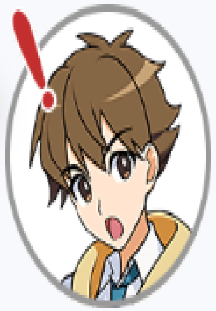
- When a program entered in the editor is executed, the result is output to the console.
- To output a character to the console, enter 'console.log("character");'.
- If there is a mistake in the program, an error will be displayed.

Got it

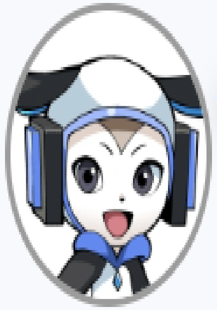
Key Points of the Lesson

To leave a note in the program

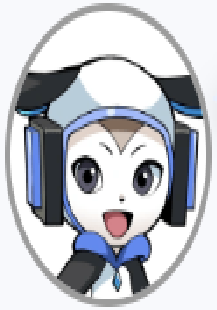

JavaScript
1 When executed, the text is output to the console.
2 console.log("Hello");
Console
Uncaught SyntaxError: Unexpected identifier 'ex
Run Code
You made an error.



Hey, I got an error message...
Ah! Come to think of it, I think I've seen a similar error before?

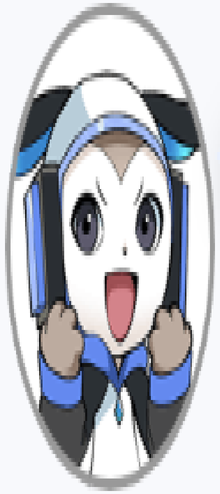


Do you remember?
We talked about how programming uses a “programming language," a special language that computers can also understand.



The note that you wrote on the first line, Rick, was an error because it was a normal sentence and not written in a way that the computer could understand it.

I see... Okay, I'll keep my notes in my notebook...



Don't worry! In fact, by **adding "//" (two slashes) at the beginning of a line**, you can leave a proper note in the program without raising an error. Let's try it!

JavaScript

```
//When executed, the text is output to the console.
console.log("Hello");
```

GREAT

Run Code ▶

Console

```
Hello
```

✓ I don't see any output for the first line of text.

Summary of this lesson

- Leaving notes in a program is called "**commenting**".

Two "/" (slashes) are placed at the beginning of each line.

JavaScript

```javascript
// When excuted., the text is output to the sonsole
console.log("Hello");
```

Check the Key Points

Let's recap what you have learned in this lesson. Check the items below and click the "Got it" button.

- [x] Leaving a note in a program is called a “comment,” and is preceded by "//" (two slashes) at the beginning of each line.

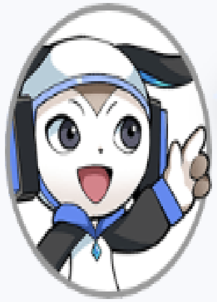
Got it

Back to Previous Lesson

Continue to Next Lesson

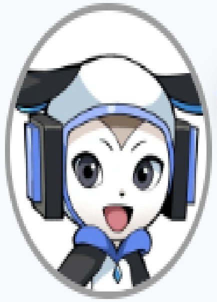
Key Points of the Lesson

About comment out



Today we're going to practice the "comments" we learned in the previous lesson!

Comment... You just enter "//" (two slashes) at the beginning of a line, right?



You remember it well! Let's give it a try!

LET'S TRY!

Try again

1 Type in the editor!

Create your program as follows.

(1) Make a comment with the sentence "Greetings for the day".

Hints [View Hints, if you're unsure.]

Sample: Output

```
Good morning
Hello
Good evening
```

JavaScript

```
//Greetings for the day.
console.log("Good morning");
console.log("Hello");
console.log("Good evening");
```

Run Code ▶

Console

```
Good morning
Hello
Good evening
```

✓ You see the output, "Good morning, Hello, Good evening."

LET'S TRY!

2 Type in the editor!

Let's modify the second line to the same program as below.

```
//console.log("Apple");
```

Hints View Hints, if you're unsure.

Sample: Output

Rerun

```
Eggs
Milk
```

JavaScript

```javascript
console.log("Eggs");
//console.log("Apple");
console.log("Milk");
```

Run Code ▶

Console

```
Eggs
Milk
```

✓ I see that "Eggs" and "Milk" have been output

Question

Does everything sound good so far?

- Yes, perfect.
- Hmmm... I'm a little unsure.

Next

Summary of this lesson

◆ If you prefix a program with //, the program is temporarily not executed.

JavaScript

```javascript
console.log("Saury");
//console.log("Apple");
console.log("Bonito");
```

Console

```
Saury
Bonito
```

Check the Key Points

Let's recap what you have learned in this lesson. Check the items below and click the "Got it" button.

- [x] You can temporarily prevent the execution of a program by prefixing it with "//".

Got it

Back to Previous Lesson

Go to Review

Review

Mini-test with hints.

Diamonds earned per correct answer

A total of 6 ques.

6 q.	◆◆◆◆◆
5 q.	◆◆◆◆◇
4 q.	◆◆◆◇◇
3 q.	◆◆◇◇◇
2 q.	◆◇◇◇◇

Subject area

console.log

Start from the beginning

Question 1

Select the correct program to output "Hello" to the console.

1
```
console.logHello;
```

2
```
console.log("Hello");
```

3
```
consolelog("Hello")
```

4
```
log.console("Hello");
```

Question 1

Select the correct program to output "Hello" to the console.

1
```
console.logHello;
```

2
```
console.log("Hello");
```

3
```
consolelog("Hello")
```

4
```
log.console("Hello");
```

View Hints

Answer

A program that outputs a character to the console is "console.log("character");".

1	✗ There are no round brackets and double quotation marks after console.log.
2	✓ Correct !
3	✗ There is no dot between console and log.
4	✗ The console and log before and after the dot are reversed.

Next

Question 2

Create a program as follows.

(1) Output "Good morning" to the console.

Text copy : Good morning

JavaScript

```
console.log
```

Reset the code

Run Code ▶

Console

View Hints

Answer

Question 2

Create a program as follows.

(1) Output "Good morning" to the console.

Text copy : Good morning

JavaScript

```
console.log("Good morning");
```

Reset the code

Run Code ▶

Console

```
Good morning
```

View Hints

Answer

Ques. 2 Expl.

✓ Correct

A program that outputs a character to the console is "console.log("character");".

JavaScript

```javascript
console.log("Good morning");
```

```javascript
console.log("Good morning");
```

Your Code

Answer Code

Next

Question 3

Create a program as follows.

(1) Output "Programming" to the console

Text copy : Programming

JavaScript

```
1
```

Reset the code

Run Code

Question 3

Create a program as follows.

(1) Output "Programming" to the console

Text copy : Programming

JavaScript

```
console.log("Programming");
```

Reset the code

Run Code ▶

Console

```
Programming
```

View Hints

Answer

✓ Correct

A program that outputs a character to the console is "console.log("character");".

JavaScript

```
console.log("Programming");
```

Your Code

```
console.log("Programming");
```

Answer Code

Next

Question 4

Create a program as follows.

(1) Output "Hello" to the console.

Text copy : Hello

JavaScript

```
1
```

Reset the code

Run Code

Question 4

Create a program as follows.

(1) Output "Hello" to the console.

Text copy : Hello

JavaScript

```
console.log("Hello");
```

Reset the code

Run Code ▶

Console

```
Hello
```

View Hints

Answer

✓ Correct

A program that outputs a character to the console is "console.log("character");".

JavaScript

```
console.log("Hello");
```

Your Code

```
console.log("Hello");
```

Answer Code

Next

Question 5

Please select the correct way to write your comment.

1 /Write your comments here.

2 //Write your comments here.

3 Write your comments here.//

4 /Write your comments here./

Question 5

Please select the correct way to write your comment.

1 /Write your comments here.

2 //Write your comments here.

3 Write your comments here.//

4 /Write your comments here./

View Hints

Answer

To comment, type "//" (two slashes) at the beginning of the line you wish to comment on.

1	✗ Slashes are at the beginning of the sentence, but only one is attached.
2	✓ Correct !
3	✗ Slashes are at the end of the sentence.
4	✗ Slashes are at the beginning and end of sentences.

Next

Question 6

Comment out the program so that the output is as follows.

```
Hello.
My name is
Cody.
```

JavaScript

```
console.log("Hello.");
console.log("My name is");
console.log("Rick.");
console.log("Cody.");
```

Reset the code

Run Code

Question 6

Comment out the program so that the output is as follows.

```
Hello.
My name is
Cody.
```

JavaScript

```javascript
console.log("Hello.");
console.log("My name is");
//console.log("Rick.");
console.log("Cody.");
```

Reset the code

Run Code ▶

Console

```
Hello.
My name is
Cody.
```

View Hints

Answer

To comment out a program, type "//" (two slashes) at the beginning of the program.

JavaScript

```javascript
console.log("Hello.");
console.log("My name is");
//console.log("Rick.");
console.log("Cody.");
```

Your Code

```javascript
console.log("Hello.");
console.log("My name is");
//console.log("Rick.");
console.log("Cody.");
```

Answer Code

Go to Results

PERFECT

6 q. / 6 q.

[Chapter1]Introduction

1 2 3 4 5 6 Review

Go to Results

PERFECT 6 q. / 6 q.

Please read the explanations carefully and review the lesson if necessary. Keep going!

1	✓	► View explanation	
2	✓	► View explanation	
3	✓	► View explanation	
4	✓	► View explanation	
5	✓	► View explanation	
6	✓	► View explanation	

Go to Next Chapter